Qwen

Architectural Blueprint for the Aegis Interview Room: A Hybrid Edge-Cloud Framework for Secure, Biometrically-Proctored Remote Assessments

Product Requirements Document (PRD)

This document defines the vision, scope, objectives, and detailed requirements for the Aegis Interview Room, an advanced AI-powered platform designed to provide a secure, frictionless, and scientifically validated remote interviewing experience. The primary goal is to create an environment that is effectively impervious to cheating while ensuring a positive and fair experience for all candidates. The system leverages a suite of multi-modal behavioral biometrics, including real-time eye tracking, advanced gesture recognition, and sophisticated anti-spoofing techniques, to achieve this objective [4] [6]. The PRD serves as the foundational blueprint for product management, engineering, and design teams, establishing a clear and unambiguous target for development. It outlines the system's core capabilities, its operational constraints, and the success metrics against which it will be evaluated.

The central vision of the Aegis Interview Room is to establish the industry standard for secure remote assessment by completely neutralizing common cheating vectors, including impersonation, ghostwriting, unauthorized assistance, and the use of hidden materials [47]. This is achieved not through invasive or punitive measures, but through a seamless integration of intelligent proctoring that respects candidate privacy and maintains a professional atmosphere [48]. The system's design is predicated on the principle of transparency; candidates must understand what is being monitored and why, fostering trust and cooperation rather than suspicion [59]. The ultimate objective is to shift the focus from manual supervision to intelligent, automated oversight, providing recruiters with actionable insights that augment their judgment rather than replacing it [100].

The primary business and technical objectives of the Aegis platform are quantifiable and ambitious. First, the system must achieve a spoofing attack detection rate of greater than 98%. This includes defeating a range of threats from simple photo attacks to

sophisticated 2D masks, 3D masks, and video replay attempts . Second, the platform aims to maintain a false-positive cheating flag rate below 2% through the use of multi-modal data fusion and contextual AI analysis, ensuring that genuine candidate behavior is not unfairly penalized . Third, to make the system's output accessible and useful, the platform must provide recruiters with an intuitive, actionable "Integrity Score" that distills complex telemetry into a single, understandable metric, supplemented by timestamped risk flags for deeper investigation . Finally, and most critically, the system must operate within a strict legal and ethical framework, achieving 100% compliance with global biometric privacy laws, including the General Data Protection Regulation (GDPR), the Illinois Biometric Information Privacy Act (BIPA), and the California Consumer Privacy Act (CCPA) [115]. These objectives form the core pillars upon which all subsequent requirements are built.

The scope of the initial release of the Aegis Interview Room is defined as web-based. The platform will consist of a candidate-facing interview portal, a real-time computer vision proctoring engine, a recruiter-facing analytics dashboard, and a secure backend for storing video artifacts and session data . The system will focus exclusively on integrity and proctoring; it will not include an automated scoring mechanism for the content of the candidate's answers. This decision allows the team to concentrate on building a robust and reliable proctoring engine before extending into the more complex domain of answer evaluation. The out-of-scope items for this phase explicitly exclude native mobile applications. While mobile access is desirable, a web-only approach is prioritized to maintain tighter control over the browser environment, ensuring that the necessary technologies like WebRTC and WebGPU are consistently available and behave predictably across different devices . Other future enhancements, such as automated scoring of interview responses, are acknowledged as potential Phase 2 features but are not part of the current development plan .

The functional requirements detail the specific behaviors the system must exhibit to fulfill its purpose. FR1 mandates Active Liveness Detection, where the system prompts the candidate to follow randomized on-screen trajectories. This action validates pupillary response and other micro-movements to definitively distinguish a live person from a static image or a pre-recorded video . FR2 requires Continuous Passive Eye Tracking, which involves monitoring gaze vectors, fixation duration, and saccades throughout the interview. The system will automatically flag prolonged off-screen gazes (e.g., exceeding three seconds) or erratic pupil dilation patterns that may suggest the candidate is reading from hidden notes or another device . FR3 specifies Advanced Gesture & Pose Recognition, utilizing skeletal tracking algorithms like MediaPipe OpenPose to detect secondary persons entering the frame, hand-to-ear gestures indicative of hidden earpieces, or any manipulation of unauthorized objects visible to the camera [95]. FR4 is

Environmental & Device Scanning, which uses object detection models to identify the presence of secondary monitors, mobile phones, or other suspicious devices in the candidate's physical environment [20]. FR5 covers Audio Forensics, involving the detection of secondary voices, keyboard typing that doesn't align with speech cadence, or anomalies in the audio stream that could indicate playback . Finally, FR6 dictates the Integrity Scoring Engine, which aggregates all telemetry from the aforementioned functions into a single 0-100 "Integrity Score" and generates a detailed timeline of risk events for the recruiter's review .

Non-functional requirements define the quality attributes and operational constraints of the system. NFR1 establishes a strict latency requirement for the client-side CV inference engine, mandating that it operates with less than 200 milliseconds of latency to prevent noticeable desynchronization between the candidate's actions and the on-screen interface . NFR2 sets high accuracy targets: facial recognition for identity verification must exceed 99.5%, and activity recognition models must achieve over 92% accuracy . NFR3 addresses the paramount concerns of privacy and security. All biometric data must be encrypted using AES-256 at rest and TLS 1.3 in transit. Personally Identifiable Information (PII) must be masked within analytics dashboards, and data retention policies must be strictly enforced, with all raw session data auto-purging after 90 days unless legally required to be retained [22]. NFR4 focuses on accessibility, requiring the user interface to comply with WCAG 2.1 Level AA standards. Furthermore, the system must provide alternative interaction flows for candidates with documented motor or visual impairments, such as the ability to disable strict gaze tracking if medically justified .

The following table details the user stories and their corresponding acceptance criteria, translating functional requirements into concrete scenarios from the perspective of the primary users.

ID	User Story	Acceptance Criteria
US-01	As a candidate, I want to verify my camera and lighting before the interview starts so that the AI can calibrate properly.	The system runs a 10-second calibration sequence that checks ambient lighting levels (lux), verifies the visibility of facial landmarks, and confirms proper camera focus before enabling the "Start" button.
US-02	As a candidate, I want clear instructions if my face leaves the frame so I can correct it without being unfairly flagged.	If the candidate's face exits the designated tracking area, a gentle, non-punitive UI overlay appears with text prompting them to "Please center your face." Only after a second warning or a prolonged absence is a formal event logged.
US-03	As a recruiter, I want to see a timeline of "Risk Events" during the interview so I can quickly review suspicious moments.	The recruiter dashboard displays a scrub-able video timeline with red markers indicating high-risk events (e.g., "Off-screen gaze > 5s", "Secondary voice detected"). Clicking a marker instantly navigates the video player to that timestamp.
US-04	As a system admin, I want the system to automatically reject video-replay attacks.	The system employs multiple techniques to detect replay, including analyzing for Moiré patterns caused by screen pixel refresh rates and performing pixel-level noise analysis to identify inconsistencies present in pre-recorded video feeds but absent in live streams.
US-05	As a candidate with a disability, I need an accommodation to allow me to participate in the interview.	During account setup, the candidate can request accommodations. The recruiter is then given a "Relaxed Proctoring" toggle in the command center, which disables biometric tracking (gaze, micro-expressions) while retaining identity and environmental checks.
US-06	As a recruiter, I want to see exactly what the AI saw when it flagged an event.	When reviewing a flagged event on the timeline, the recruiter can activate a "Ghost View" mode. This overlays the live video feed with the AI's perception data, such as the MediaPipe skeletal wireframe and gaze vectors, allowing direct correlation between the flag and the visual evidence.

These documents collectively form the authoritative source for the Aegis Interview Room project, providing a complete and unambiguous definition of what the system is, what it must do, and how its success will be measured.

Technical Requirements Document (TRD)

The Technical Requirements Document (TRD) provides a deep dive into the architectural decisions, technology stack, and performance specifications required to build the Aegis Interview Room. This document is intended for the engineering and infrastructure teams, serving as the definitive guide for implementation. It details the system's logical and physical structure, the APIs that enable communication between components, and the rigorous standards governing performance, security, and compliance. The architecture is deliberately designed to balance the competing demands of low-latency real-time processing, massive scalability, and stringent data privacy, forming the technical foundation upon which the entire platform rests.

The cornerstone of the Aegis architecture is a **Hybrid Edge-Cloud Model**. This design choice is critical to meeting the system's demanding low-latency requirements for real-time computer vision tasks [74] [91]. Heavy, latency-sensitive inference operations—such as facial landmark detection, gaze estimation, and gesture recognition—are executed locally on the candidate's client device within the browser environment. This client-side processing is made possible through the use of WebAssembly (Wasm) and WebGPU, ensuring that computationally intensive tasks do not rely on slow round-trip communications to a distant server . By processing data at the edge, the system minimizes network latency, preventing stream desync and providing a responsive user experience [75]. In contrast, the cloud backend is responsible for orchestrating sessions, ingesting the high-throughput stream of anonymized telemetry data, performing heavy-duty analytics and long-term storage, and powering the recruiter's analytics dashboard [103]. This division of labor optimizes both performance and cost, leveraging the computational power of the client device for immediate tasks and the vast scalability of the cloud for aggregate analysis and archival [93]. The architecture is inherently modular and forward-compatible, allowing for the independent upgrade of ML models on either the edge or cloud side without disrupting the entire system [94].

The client-side, or Edge AI, component is the first line of defense and intelligence. It relies on several key technologies. **WebRTC** is used for capturing high-fidelity, low-latency audio and video streams directly from the candidate's webcam and microphone [2] [77]. For the inference engine, **TensorFlow.js** or **ONNX Runtime Web** will be employed to run quantized machine learning models optimized for browser environments [50]. These models include: a lightweight CNN for **Gaze Estimation**, based on frameworks like MPIIGaze or approaches described in recent literature, designed for real-time performance [23] [24]; a **Pose Estimation** model like MoveNet or MediaPipe Pose to track body posture ; and a specialized **Liveness** model that analyzes pupillary light reflex changes in response to screen brightness shifts to counter spoofing . The use of ONNX Runtime is particularly strategic, as it provides a vendor-agnostic, high-performance inference engine capable of running on various backends, including WebGPU for GPU acceleration, and supports model formats from multiple training frameworks [105][107].

The server-side, or Cloud Backend, is a scalable, containerized environment built on **Kubernetes** to manage a cluster of microservices . Communication with the backend is managed through a GraphQL API for flexible data fetching in the frontend application and RESTful webhooks for event notifications. High-volume, real-time telemetry data streamed from clients is ingested via an **Apache Kafka** cluster, which decouples the data producers (clients) from the consumers (processing workers) and ensures no data loss even under peak load . This data is then processed by Python/FastAPI microservices

running on GPU-enabled nodes (e.g., NVIDIA A10G) for tasks that require more computational power, such as complex anomaly detection algorithms or forensic analysis of flagged sessions . Video recordings and other large artifacts are securely stored in **AWS S3** buckets, which are configured for private access only, with file access granted temporarily via short-lived, pre-signed URLs . This architecture ensures high availability (targeting 99.99% uptime) and fault tolerance, critical for a service used in high-stakes assessments .

The hardware and software requirements are specified for both the client and server sides. On the client side, candidates must have a minimum of a 720p webcam, a dual-core CPU equivalent to an Intel i5 or AMD Ryzen 5, and 8GB of RAM to ensure smooth operation of the local CV engine . The software environment requires a modern evergreen browser (Chrome, Edge, Safari) with support for WebGPU/WebGL and a stable internet connection with a minimum upload speed of 5 Mbps and jitter below 100ms . Server-side requirements include a Kubernetes cluster provisioned in a major cloud provider, with auto-scaling groups to dynamically adjust compute resources based on demand. GPU nodes are reserved for the processing workers to handle fallback inference or intensive post-session analysis . Storage is handled by AWS S3 for media and a combination of PostgreSQL for relational data and TimescaleDB for time-series telemetry .

The API design is centered around WebSocket for real-time communication. The `/api/v1/session/init` endpoint is responsible for generating secure, time-limited tokens for WebRTC connections and initializing the CV models on the client side. The most critical endpoint is `/ws/v1/telemetry`, a dedicated WebSocket channel where the client continuously streams compressed JSON payloads containing biometric data points such as `[timestamp, gaze_vector, head_pose_euler, gesture_confidence]` . To manage bandwidth, these payloads must be serialized using a compact format like MessagePack or Protobuf, with a target throughput kept under 50 KB/s per session . Finally, the `/api/v1/integrity/report` endpoint aggregates all session data from the database and generates the final, human-readable report for the recruiter .

Performance benchmarks are non-negotiable. The CV models must process video at a minimum of 30 frames per second (FPS). If the FPS drops below 15, the client-side application must alert the user to close background applications to free up resources . The backend's Apache Kafka pipeline must be able to handle the aggregate telemetry load from thousands of concurrent interviews without data loss. The cloud API gateway must maintain 99.99% uptime to ensure the service is always available .

Security and compliance are woven into every layer of the technical design. The system must adhere to the strictest standards for biometric data protection, including GDPR, BIPA, and CCPA [115]. All biometric data, including facial geometry templates derived from embeddings, must be hashed before being stored in the relational database, with only the hash value retained to prevent reverse-engineering . Raw biometric data is never stored in plain text. All data transmission is secured with TLS 1.3, and all video files are encrypted at rest using AWS S3's server-side encryption with KMS-managed keys (SSE-KMS) . The system must implement robust anti-spoofing measures, including detecting Moiré patterns and audio-visual sync drift to thwart video replay attacks, and analyzing micro-expressions and blood flow via remote photoplethysmography (rPPG) to validate the presence of a live heartbeat, thereby countering deepfake videos [71] . Access control is managed through fine-grained IAM roles, and all administrative actions should be logged in an immutable audit trail to ensure accountability [111].

Application Flow

The Application Flow delineates the complete, end-to-end journey of a candidate participating in an Aegis Interview Room session, from initial onboarding to the final delivery of the integrity report to the recruiter. This flowchart is meticulously designed to integrate the platform's advanced proctoring capabilities seamlessly into the user experience, ensuring maximum security without introducing undue friction. The process is divided into three distinct phases: Pre-Interview Onboarding & Environment Calibration, Interview Execution & Continuous Monitoring, and Post-Interview & Reporting. Each phase contains a series of logical steps and checkpoints, with a particular emphasis on the moments where the AI-driven anti-cheating features are activated and validated.

Phase 1: Pre-Interview Onboarding & Environment Calibration

This initial phase is critical for setting the stage for a secure and successful interview. Its primary goals are to authenticate the candidate, verify their equipment, scan their environment for potential threats, and calibrate the AI models for their specific physical context.

1. **Authentication:** The process begins when a candidate receives a secure, time-limited magic link via email. They log in using a Single Sign-On (SSO) provider or a

secure one-time password (OTP) sent to their registered email address. This step ensures that only authorized individuals can access the system .

2. **Hardware Check:** Upon logging in, the system immediately initiates a hardware check. It requests permission to access the candidate's webcam and microphone, and simultaneously runs a background test to measure the available internet upload speed and network jitter. The system will display warnings if the hardware or network conditions fall below the minimum requirements (e.g., upload speed < 5 Mbps), giving the candidate an opportunity to resolve the issue before proceeding .
3. **Environment Scan (360°):** To ensure the candidate is alone in a controlled environment, the system instructs them to slowly pan their camera around the room. Using a real-time object detection model like YOLOv8, the system scans the video feed for the presence of secondary monitors, mobile phones, or other unauthorized persons [20]. If an object of concern is detected, the system pauses and highlights the item in the candidate's view, asking them to confirm its legitimacy or remove it from the environment.
4. **Active Liveness Challenge:** Before the main calibration, the system executes an active liveness check to definitively prove the candidate is a live person. The screen displays a small, randomly moving dot. The candidate is instructed to follow the dot with their eyes only, without moving their head. The system analyzes the resulting eye movements, pupillary response, and subtle 3D depth mapping to differentiate between a live human and a flat photo, mask, or video replay . This checkpoint is a crucial defense against spoofing attacks.
5. **Baseline Calibration:** Once liveness is confirmed, the system performs a final baseline calibration. The candidate is asked to stare straight at the camera for five seconds. During this period, the system records baseline metrics such as ambient lighting levels, skin tone characteristics, resting pupil diameter, and the candidate's natural gaze point relative to the camera lens. This baseline data is later used to normalize and interpret deviations during the actual interview, improving the accuracy of anomaly detection .

Phase 2: Interview Execution & Continuous Monitoring

With the candidate successfully onboarded and calibrated, the interview commences. This phase is characterized by the continuous, passive operation of the AI proctoring engine, which runs in parallel with the interview itself.

1. **Question Delivery:** The interviewer, who can be a human recruiter or an AI avatar, delivers the first question. This can be done via a live video feed or by displaying a text prompt on the screen . The candidate's own Picture-in-Picture (PiP) feed of

themselves is displayed in the bottom-right corner of the screen, a standard feature to help them feel more comfortable .

2. **Passive Telemetry Streaming (Continuous):** Throughout the interview, the client-side CV engine continuously streams high-frequency telemetry data to the cloud backend via a WebSocket connection . This data forms the basis for all real-time analysis:

- **Eye Tracking:** The system constantly monitors gaze vectors, fixation points, and saccades. A *checkpoint* is triggered if the candidate's gaze remains in the top-right portion of their screen for more than three seconds—a common location for hidden notes or a second monitor. This raises a low-level flag for the integrity engine [29].
- **Head Pose:** Head pose is tracked in 3D (Yaw, Pitch, Roll). A *checkpoint* is triggered if the head turns away from the primary camera by more than 45 degrees. In response, the system may mute the candidate's microphone and display a gentle warning overlay, prompting them to face the camera .
- **Gesture Tracking:** The MediaPipe Hands model tracks the position of the candidate's hands. A *checkpoint* is raised if the hands leave the field of view or approach the ears, signaling a potential hidden earpiece [11].

1. **Randomized Active Checks:** To prevent deepfake or video injection attacks, the system performs randomized active checks every 5 to 10 minutes. These can be subtle, such as briefly flashing the screen with a specific color pattern, or interactive, asking the candidate to perform a quick micro-action like "blink twice" or "nod once" . These checks re-verify the candidate's liveness and ensure the stream has not been replaced.
2. **Audio Forensics:** Simultaneously, a background process analyzes the audio stream. A *checkpoint* is triggered if the system detects keyboard typing sounds that are not synchronized with the candidate's speech rhythm, or if it identifies whispers or speech from a secondary, unidentified voice source .

Phase 3: Post-Interview & Reporting

Once the interview concludes, the system enters a final phase focused on data aggregation, analysis, and report generation.

1. **Session Termination & Log Finalization:** When the candidate ends the session, the live video stream is terminated. The client-side CV engine finalizes the local telemetry log, compresses it, and uploads the summary data packet to the cloud backend for processing .
2. **Data Aggregation & Analysis:** On the cloud side, a processing worker takes the raw telemetry data (gaze, head pose, etc.) and the recorded video. It applies the

"Integrity Scoring Algorithm," which cross-references timestamps from the video with the telemetry data to identify and score all risk events that occurred during the session .

3. **Recruiter Dashboard Generation:** The results are compiled into a comprehensive report and loaded into the recruiter's Command Center dashboard. The video is indexed and overlaid with telemetry visualizations, including a gaze heatmap showing where the candidate looked most often and a timeline of all detected risk events [31] . The final Integrity Score is calculated and displayed prominently .
4. **Recruiter Review:** The recruiter can now review the session. They watch the indexed video and can click on any red-flagged event on the timeline. This action instantly jumps the video player to that exact moment and pauses it, bringing up a detailed panel that shows the precise CV data that triggered the flag (e.g., "Gaze vector detected looking at secondary screen for 4.2 seconds") . This allows the recruiter to make a contextual, informed judgment about the candidate's conduct, acting as the ultimate arbiter in the process [100].

This structured application flow ensures that every aspect of the interview is monitored and analyzed by the Aegis system, creating a robust and defensible record of the candidate's behavior while maintaining a logical and transparent process for all parties involved.

UI & UX Design Flow

The User Interface (UI) and User Experience (UX) design for the Aegis Interview Room is guided by a core philosophy centered on **minimalist cognitive load reduction** and **transparent proctoring**. The design choices are deliberate and rooted in psychological principles aimed at mitigating two primary risks: triggering false positives in the AI systems due to candidate anxiety and nervousness, and creating a negative candidate experience that could damage the employer's brand [129][130]. Every element of the interface, from the placement of controls to the style of feedback messages, is crafted to be unobtrusive, intuitive, and supportive, ensuring that the proctoring technology works in harmony with, rather than against, the human participants.

For the **Candidate Interface (The Interview Room)**, the design is intentionally sparse and clean. The visual environment is curated to minimize distractions and reduce the candidate's cognitive burden, which helps produce more natural and authentic behavior for the AI to analyze [130][131]. The screen layout is carefully composed: the primary content

—the interviewer's video feed or the question text—is placed centrally, commanding the user's full attention. In the bottom-right corner, a small Picture-in-Picture (PiP) window displays the candidate's own webcam feed, providing a sense of presence and helping them monitor their own on-camera appearance without needing to look away . The top bar of the interface contains only essential, non-distracting information: a minimalist timer, an indicator for internet connection strength, and a subtle badge informing the candidate that their session is being recorded and proctored . During the initial calibration phase, a faint, semi-transparent oval guide appears to gently nudge the candidate into the optimal head position, but this guide fades away entirely once the interview begins to avoid constant visual clutter .

The UX micro-interactions are designed to be empathetic and non-punitive. The system's warning mechanisms are a prime example. Instead of a jarring, red-alert-style pop-up that would startle the candidate and cause erratic movements, the system uses a soft, glowing amber border around the screen, accompanied by calm text: "*Please keep your face within the frame.*" This provides clear guidance without inducing panic . Similarly, when an active liveness check is triggered, a smooth, non-intrusive modal slides in from the side, stating, "*For security, please follow the dot with your eyes,*" framing the request as a necessary step for the interview process, not a punitive action . This gentle feedback loop is crucial for maintaining a positive candidate experience, which is a key factor in attracting top talent [33] .

On the other side of the proctoring equation is the **Recruiter / Proctor Interface (The Command Center)**. This dashboard is designed for efficiency and deep analytical insight. Its layout is divided into two main panels: a large left panel for the video player and a right panel for analytics . The video player displays the recorded interview, but it is enhanced with powerful analytical tools. The right panel features a large, circular gauge prominently displaying the candidate's final "Integrity Score" (e.g., 94/100 - "High Integrity"), providing an immediate, at-a-glance assessment . Below the score is a multi-layered telemetry timeline that overlays analytical data onto the video [31] . This timeline might include layers for Gaze Heatmap (showing areas of visual focus), Head Pose Graph (visualizing head movements), and Audio Anomalies (highlighting spikes in background noise or typing) [61] . At the bottom of the screen, an AI-generated transcript of the interview is displayed, with keywords potentially highlighted to draw the recruiter's attention to specific topics .

Several key UX features empower the recruiter to investigate flagged events effectively. The "Ghost View" is perhaps the most important. By toggling this mode, the recruiter can superimpose the AI's perception data directly onto the video feed. This means they can see the MediaPipe skeletal wireframe drawn over the candidate's body and the precise

gaze vectors emanating from their eyes, allowing them to visually correlate the AI's algorithmic findings with the actual video footage . Another critical feature is "Quick Jump." When a recruiter clicks on a red flag icon on the telemetry timeline, the video player instantly navigates to that exact timestamp and pauses, presenting the detailed telemetry data that prompted the flag in a side-panel . This transforms the review process from a tedious, frame-by-frame search into a targeted investigation of specific events.

Finally, the design is fully committed to **Accessibility and Usability**. The entire candidate interface must be navigable using only a keyboard (`Tab` for focus, `Enter` to select), catering to users who cannot operate a mouse . Screen reader support is provided through appropriate ARIA labels for all dynamic status updates related to the proctoring system . Most importantly, the system includes an "Accommodation Mode." If a candidate discloses a medical condition, such as a tremor, ADHD, or a disability that affects eye contact, the recruiter can activate this mode. This toggles a setting that disables the more sensitive biometric tracking modules (like gaze and micro-expression analysis), relying instead on a more basic suite of checks focused on identity verification and environmental scanning. This ensures the platform is inclusive and compliant with accessibility standards, treating all candidates fairly while still upholding the core security objectives .

Backend Schema & Data Architecture

The backend data architecture for the Aegis Interview Room is designed for scale, performance, and security, employing a **polyglot persistence** strategy. This approach recognizes that different types of data generated by the system have vastly different characteristics and access patterns, making a single database technology suboptimal. Relational databases are used for structured metadata, time-series databases for high-frequency telemetry, object storage for large binary blobs like video, and vector databases for similarity searches on biometric data . This specialization ensures that each data type is stored in the most efficient and effective manner possible, supporting the platform's demanding real-time and analytical workloads while adhering to strict privacy and compliance requirements.

The relational data, which includes user profiles, session metadata, permissions, and billing information, is managed by a **PostgreSQL** database . The core schema for this relational store is designed to be normalized yet efficient. The `users` table stores candidate, recruiter, and admin profiles, with a crucial column for

`facial_embedding_hash`. This column stores a cryptographic hash of the candidate's facial embedding vector, not the vector itself, providing a strong layer of security that prevents the reconstruction of the original biometric template from the database . The `sessions` table links candidates to recruiters, tracks the status and timing of each interview, and stores the final aggregated `integrity_score` . To log individual risk events, a `risk_events` table is created. This table is linked to a session and contains a `timestamp_ms`, the `event_type` (e.g., 'gaze_offscreen', 'phone_detected'), a `severity` level, a `confidence_score`, and a `metadata` field in JSONB format. The JSONB column is ideal for storing the diverse and nested telemetry data associated with each specific event, such as the exact gaze vector coordinates or the bounding box of a detected object .

For the high-frequency, time-series data generated by the CV engine—such as gaze pitch/yaw, head pose, and pupil diameter sampled at 30-60fps per session—the architecture utilizes a **Time-Series Database** like TimescaleDB (an extension for PostgreSQL) or InfluxDB . This specialized database is optimized for rapid writes and efficient querying of time-stamped data. The `biometric_telemetry` table is configured as a hypertable, which automatically partitions the data into chunks based on time. This partitioning dramatically improves query performance when analyzing telemetry for a specific session or time window, as the database engine can operate on a much smaller subset of the data rather than scanning the entire table . The schema for this table is straightforward, with a `time` column as the primary index, a `session_id` foreign key, and columns for each biometric measurement (e.g., `gaze_pitch`, `head_roll`) . This structure is perfectly suited for the massive write throughput required during an interview and for the analytical queries needed to generate heatmaps and other temporal visualizations post-interview [82] .

Large, unstructured binary data, primarily the recorded video and audio files from interviews, are stored in a highly durable and scalable **Object Storage** system, such as **AWS S3** . This approach separates the data from the application logic, allowing for easy backup, versioning, and distribution. Crucially, these S3 buckets are configured to be private by default. Access to any video file is granted not through direct links but via pre-signed URLs generated by the backend. These URLs are configured with a very short expiration time (e.g., 15 minutes), ensuring that even if leaked, they cannot be used to gain long-term access to sensitive interview recordings . Furthermore, all data written to S3 is encrypted at rest using server-side encryption with keys managed by AWS Key Management Service (S3 SSE-KMS), providing an additional layer of security .

To support advanced identity verification and anti-spoofing techniques, the architecture incorporates a **Vector Database**, such as Pinecone or Milvus . The facial recognition

system generates a high-dimensional vector representation (embedding) of the candidate's face during the initial liveness check. This vector captures the unique geometric features of their face. Instead of storing these vectors in a traditional RDBMS, they are stored in the vector database, which is optimized for performing fast k-nearest neighbor (k-NN) searches. When a candidate logs in, their live facial embedding is compared against the stored baseline embedding to verify their identity. This same vector database can also be used to detect morphing attacks by comparing the statistical properties of a live face against known morphed-face patterns [89].

The data security and encryption strategy is comprehensive and multi-layered. **PII Masking** is applied at the application layer; for instance, candidate names and emails are encrypted before being inserted into the PostgreSQL database . The principle of **Privacy by Design** is embedded in the architecture, with a strong emphasis on **data minimization**. By deploying a hybrid edge-cloud model, the system processes and analyzes the majority of biometric data locally on the candidate's device. Raw, identifiable video streams are not uploaded to the cloud; only anonymized telemetry packets are transmitted, significantly reducing the privacy risk [92] [116]. The data retention policy is automated and rigorously enforced. Cron jobs or AWS Lambda functions are scheduled to periodically scan the database and object storage. After a session's data has served its purpose for review and analytics (typically 90 days), the corresponding entries in the `biometric_telemetry` table, the video files in S3, and the records in the `risk_events` table are permanently deleted [22]. This automated purging is a key requirement for demonstrating compliance with regulations like GDPR, which mandate that personal data should not be kept longer than necessary [81]. For critical administrative actions, such as a recruiter viewing a flagged session, an immutable audit log is maintained, potentially using a Merkle Tree pipeline to ensure the integrity of the log itself [99] [111].

Implementation Plan & Risk Management

The implementation of the Aegis Interview Room is structured as a six-month, five-phase roadmap leading to General Availability (GA). This plan outlines a systematic progression from initial discovery and prototyping to full-scale deployment, incorporating key milestones, resource allocation, and a proactive approach to risk management. The methodology emphasizes iterative development, continuous testing, and early engagement with legal and security experts to mitigate potential pitfalls.

Phase 1: Discovery & Core ML Prototyping (Weeks 1-6) The initial phase is dedicated to solidifying the product and technical foundations. The primary milestone is the approval of the final PRD and TRD, along with the completion of all UI/UX wireframes and mockups . Concurrently, the core Machine Learning (ML) engineers will focus on training and optimizing the lightweight CV models for deployment in a browser environment. This involves quantizing models like MediaPipe Face Mesh and gaze estimation CNNs for TensorFlow.js or ONNX Runtime Web [2] . A significant technical challenge is developing the Active Liveness algorithm, which must reliably detect pupillary light reflexes and other micro-movements in response to randomized screen stimuli . The expected deliverable for this phase is a working local prototype of the CV engine, capable of running in a browser at over 30 FPS and correctly identifying facial landmarks and basic gaze directions.

Phase 2: Backend & Infrastructure Setup (Weeks 7-12) With the front-end concepts proven, Phase 2 shifts to building the scalable and secure backend infrastructure. The first milestone is the provisioning of the cloud environment, including the Kubernetes cluster, PostgreSQL and TimescaleDB instances, and AWS S3 buckets . The next critical task is developing the WebRTC signaling servers to facilitate low-latency, peer-to-peer video streaming between the candidate and the recruiter/dashboard [77] . Following this, the team will construct the high-throughput WebSocket telemetry ingestion pipeline, using Apache Kafka to buffer incoming data from clients before it is processed and written to the time-series database . A parallel and equally critical milestone is the implementation of all security protocols, including end-to-end encryption, IAM role configurations, and the initial setup for GDPR/BIPA compliance, likely involving consultation with the dedicated Security/Compliance Specialist [115]. The deliverable for this phase is a fully functional, secure, and isolated backend environment with all core APIs and database schemas in place.

Phase 3: Full-Stack Integration & UI Development (Weeks 13-18) This phase focuses on integrating the separate frontend and backend components into a cohesive, end-to-end application. The first milestone is the development of the Candidate Interview Room UI using a modern framework like React or Next.js, with the local CV engine successfully integrated and streaming data to the backend . Simultaneously, the Recruiter Command Center dashboard is built, featuring the video player, telemetry timeline, and integrity score visualization . A key milestone is the implementation of the Integrity Scoring Algorithm, which will consume the time-series telemetry data and generate the final score and risk event log for each session . The final deliverable of this phase is an Alpha version of the complete application, ready for internal testing and refinement.

Phase 4: Adversarial Testing & Beta (Weeks 19-22) Before public release, the platform undergoes rigorous adversarial testing and a closed beta program. The first milestone is a comprehensive **Red Teaming** exercise, where external ethical hackers are hired to attempt to bypass all anti-cheating features. Their goal is to defeat liveness detection using 3D masks, replay attacks, deepfakes, and other methods. The results of these tests are used to tune and harden the CV models . The second milestone is a **Bias & Fairness Audit**, conducted by a third party to evaluate the performance of the CV models across diverse demographics, including different skin tones, genders, and ages, to ensure there is no algorithmic bias [101][127]. The final milestone is a closed beta program involving 50 friendly enterprise clients and approximately 500 candidate testers. Feedback on usability, stability, and fairness is gathered to make final adjustments. The deliverable is a hardened, audited, and beta-tested platform, considered ready for GA.

Phase 5: Launch & Scaling (Weeks 23-24) The final phase culminates in the production launch. The first milestone is the deployment to a production environment with auto-scaling enabled to handle anticipated traffic . This is followed by the finalization of all documentation, API keys, and SDKs to facilitate enterprise integrations. The project concludes with the **General Availability (GA) Launch**, marking the official public release of the Aegis Interview Room. The final deliverable is the live, fully supported platform.

Resource allocation for this six-month effort is substantial. The team consists of a Project Lead/Arcitect, three Computer Vision/ML Engineers specializing in edge inference, two Backend Engineers focused on WebRTC and data pipelines, two Frontend Engineers skilled in React and complex UIs, one UX/UI Designer, a part-time Security/Compliance Specialist, and two QA/Red Team members .

A proactive risk management plan is integral to the project's success. The following table outlines the key risks, their assessed impact and probability, and the specific mitigation strategies in place.

Risk	Impact	Probability	Mitigation Strategy
High False Positive Rate	High	Medium	Implement a "grace period" and contextual AI. If a candidate looks away, prompt them before flagging. Require mandatory human review for any score below 80 to override automated flags [100].
Client Hardware Limitations	High	High	The CV engine must have a graceful degradation "fallback mode." If WebGPU fails or FPS drops below 15, the system switches to a less intensive processing mode or relaxes the strictness of certain proctoring rules .
Privacy Backlash / Legal Challenges	Critical	Low	Engage legal counsel specializing in biometric privacy (GDPR, BIPA) from day one. Prioritize privacy by design, processing data locally on-device as much as possible to minimize data sent to the cloud [116][133].
Evolving Spoofing Threats (Deepfakes)	Critical	Medium	Treat this as an ongoing arms race. Allocate 15% of post-launch sprint capacity to research and develop new anti-spoofing techniques, such as improved rPPG (heartbeat detection) and Moiré pattern analysis [71] .
Scalability Failure Under Load	High	Medium	Use a Kubernetes cluster with auto-scaling for all stateless microservices. Employ a message queue (Kafka) to buffer telemetry data, decoupling clients from the processing backend and preventing overload .
Algorithmic Bias Leading to Discrimination	Critical	Medium	Conduct a rigorous, third-party Bias & Fairness Audit during the beta phase using diverse demographic datasets. The "Accommodation Mode" provides a legal and ethical escape hatch for affected candidates [127].

This comprehensive plan and risk assessment provide a clear path to delivering a technologically advanced, ethically sound, and commercially viable AI proctoring platform.

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